The Null Cluster as Structural Imprint

A Closed Source Task and the Null-Boundary Programme

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Abstract

Null boundaries recur in causal propagation, horizon thermality, entropy bounds, asymptotic memory and modular geometry. A common interpretation requires more than their appearance in the same list: one must identify what a null interface reads, what it omits, and which resources make it competitive with other interfaces.

We give a closed comparison on a scalar-source family $S = \theta I + \alpha g$ in a supplied Lorentzian laboratory. Equal-gain directional probes use the same Gaussian pointers and a common nine-bit readout. Null contraction removes the unknown trace term. At the specified mean-square target 1.6 it needs one record, whereas every adaptive scheme restricted to either of two non-null causal menus needs at least three; an eight-record scheme suffices. In the growing-range analogue benchmark the optimal risks differ by a factor of four. The lower bound allows biased estimators, analogue access and arbitrary computation. Mixing causal types or allowing directions sufficiently close to null removes the relevant obstruction. The null record remains blind to a genuine potential contribution to the source.

This result establishes a task-specific mechanism for null compression, not a universal null-data theorem. It motivates a broader hypothesis: null interfaces are economical where causal transport and quotienting unwanted source components coincide. We develop that hypothesis against the distinct mechanisms behind the null cluster and identify source completion, actual detector costs and propagation tails as decisive tests.

1 The question behind the cluster

Light cones, horizons and radiation at null infinity do different jobs. Yet each connects an accessible region to information that reaches its boundary. This recurrence is worth explaining. The proposed *null cluster* consists of causal reachability, horizon restriction, thermal response, entropy accounting, infrared memory, geometric modular flow in special states, and massless propagation. The facts themselves belong to established relativity and quantum field theory; the question here is whether their conjunction identifies a useful principle of physical readout.

The strongest version would say that null data are the cheapest adequate interface for a specified physical task. That statement needs a target, a reconstruction law, a detector menu and a loss function. Comparing characteristic data reconstructed with the field equation against bulk samples reconstructed without it would reward unequal knowledge, rather than the causal

type of the interface. Counting the dimensions of a surface also says little about its bandwidth, sampling or required initial and boundary data.

We therefore begin with a comparison in which all readers know the same source law and have access to the same elementary meter. The null reader wins because its coupling rejects a nuisance source component, not because its competitor is denied the equations. This gives the programme an explicit positive instance. Its limits matter equally: the component discarded as a nuisance for one task is physical source information for another.

Q-RAIF A supplies the null cone of a first-order Hermitian propagation symbol when all its bands share a nondegenerate quadratic shell [18]. In that class the spatial coefficients obey Clifford relations, and both frequency branches must be present to identify the whole null cone. A common cone for different species further requires that paper's interaction and infrared assumptions; species mixing alone is insufficient. The Lorentzian laboratory used here is an input. If it is identified with a propagation sector, its detector coordinates and gain must be calibrated to that sector's cone.

Q-RAIF C distinguishes a closed carrier's representation from the smaller exact algebra left after a coupled drive is discarded [19]. Neither result supplies the stress-to-pointer transducer in our experiment. The connection established here is what null contraction does to a declared source and measurement law. The source-completeness question belongs to the companion no-view analysis [20].

2 A common source and a common meter

2.1 The scalar-source family

Work in a fixed $1+1$ laboratory with $g = \mathrm{diag}(-1, 1)$. Choose the two-parameter stress family

$$S = \theta I + \alpha g = \begin{pmatrix} \theta - \alpha & 0 \\ 0 & \theta + \alpha \end{pmatrix}, \qquad (\theta, \alpha) \in \mathcal{B} = [0, 20] \times [-30, 0]. \tag{1}$$

The target is $\theta = (S_{00} + S_{11})/2$. The trace coefficient α is unknown and must not be supplied to any estimator.

These tensors have a simple matter realization. For a canonical scalar,

$$\mathcal{L} = -\tfrac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V, \qquad S_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi - g_{\mu\nu} \big(\tfrac{1}{2} (\partial \phi)^2 + V \big). \tag{2}$$

On the observation patch take $\phi = \sqrt{2\theta}\, t$ and the constant potential $V = -\alpha \geq 0$. The scalar equation is satisfied and (2) gives (1); at $\theta = 0$ the field is constant and the potential term remains. This is the usual scalar stress formula [1, 2], not a new matter equation. Preparation of the patch, its boundary conditions and the background geometry are common physical inputs. A homogeneous field on an infinite flat background is not being claimed as a finite isolated self-gravitating apparatus.

In this family the null energy measures the gradient contribution while discarding the constant potential. Positivity of that potential does not make it fictitious: the laboratory energy density is $S_{00} = \theta + V$. Null sufficiency for θ is consequently weaker than source completeness.

2.2 Directional permission

An elementary probe is a laboratory-Euclidean unit vector w: $w_0^2 + w_1^2 = 1$. Its signal is

$$q = g(w, w) = w_1^2 - w_0^2, \qquad s_q = w^{\mathsf{T}} S w = \theta + q\alpha. \tag{3}$$

A null vector $w = (1,1)/\sqrt{2}$ gives $q = 0$. The two non-null menus are

$$\mathcal{Q}_{\mathrm{T}} = [-1, -q_0], \qquad \mathcal{Q}_{\mathrm{S}} = [q_0, 1], \qquad q_0 = \tfrac{1}{3}. \tag{4}$$

The null menu contains $q = 0$. Either sign of the null direction gives the same signal in this diagonal source family.

The Euclidean normalization describes a calibrated detector gain in this laboratory; it is not Lorentz invariant. A proper-unit timelike or spacelike vector of rapidity η, after this normalization, has $|q| = \operatorname{sech}(2\eta)$. Thus the non-null menus can be realized with a supplied boost bound $|\eta| \le \frac{1}{2}\operatorname{arcosh} 3$. A direction reference, alignment, gain calibration and this actuator permission are part of the experiment. Rescaling a vector to increase its signal is not a free operation. The finite rapidity bound will be used in the proof; removing it changes the answer.

2.3 Actual readout and the resource boundary

Each read uses a fresh independent pointer with canonical coordinates Q, P, $[Q, P] = i\hbar$, initially in

$$\psi(Q) = (2\pi\sigma^2)^{-1/4} \exp[-Q^2/(4\sigma^2)], \qquad \sigma = 1. \tag{5}$$

The supplied interaction pulse implements

$$U_q = \exp(-i s_q P/\hbar), \qquad H_{\mathrm{int}}(t) = \chi(t) s_q P, \qquad \int \chi(t)\, dt = 1. \tag{6}$$

Position measurement of the translated pointer gives

$$Y_j = \theta + q_j \alpha + Z_j, \qquad Z_j \overset{\mathrm{iid}}{\sim} N(0,1). \tag{7}$$

This is an actual meter coupling for a stable classical source parameter, or for a supplied commuting register controlling the translation. It does not grant a nondisturbing measurement of arbitrary quantum stress. The front end that transduces (3) into the pulse, the pulse clock and the drive are supplied identically to all menus.

The finite output is a common digitizer. Clip Y to $[-64, 64]$, then round to the nearest multiple of $\Delta = 1/2$. There are 257 possible labels, stored in nine bits. Equivalently, this is a finite position-bin POVM, with the two endpoint bins extending to infinity; no intermediate infinite-precision record is required. The estimator receives only these labels. Retaining a spent pointer does not grant an additional unpriced measurement of it. The lower bound below gives its competitors the stronger permission of full analogue outcomes, so digitization cannot invalidate it.

All source means obey $-30 \le s_q \le 50$. Thus a common bounded per-read energy allowance can be assigned to every menu. For example, in dimensionless oscillator units $H_P = (P^2 + Q^2)/2$ and $\hbar = 1$, the initial mean energy is $5/8$ and the translated mean is $5/8 + s_q^2/2 \le 1250 + 5/8$. This allowance is a shared cap, not an assertion that the realized drive work is the same for every parameter and direction. Preparing the meters, converting their outcomes, communicating the labels and retaining spent pointers all cost resources. No reset or autonomous work store is hidden in (6).

For the comparison we price the number of reads under this fixed per-read interface, with a maximum pathwise stock of N pointers and $9N$ output bits. Computation may be arbitrary; it is granted to every reader, not counted as a null advantage. Choices of q_j may depend on previous actual records and on independent random seeds. Initial registers and fresh noise carry no additional information about θ, α. These conditions exclude an unpriced source oracle and allow fully adaptive competitors.

3 Sufficiency and a fair lower bound

Definition 3.1 (Closed task)**.** The task is to estimate θ in (1) with worst-case mean squared error at most $\varepsilon_* = 1.6$. A protocol is task-sufficient if it meets that bound uniformly on $\mathcal{B}$. Its read cost is the maximum number of elementary reads on any branch. Let C_{N}, C_{T} and C_{S} be the minimum costs in the three menus.

This is noisy task sufficiency, rather than exact recovery of the source tensor. In the analogue null experiment, the sample mean is a sufficient statistic for the normal location parameter θ. Our actual finite labels lose some of that information but still meet the task.

Theorem 3.2 (Adaptive single-type lower bound)**.** *For either non-null menu* (4)*, every protocol with at most N reads, including analogue readout, arbitrary randomization and biased estimation, has worst-case risk at least*

$$\sup_{\mathcal{B}} \mathbb{E}_{\theta,\alpha}(\widehat{\theta} - \theta)^2 \;\geq\; \frac{1}{4\pi^2/20^2 + N/4}. \tag{8}$$

Proof. Restrict the spacelike source family to $\alpha = -3\theta/2$, and the timelike family to $\alpha = -30 + 3\theta/2$, with $0 < \theta < 20$. Both subfamilies lie inside $\mathcal{B}$. Revealing the subfamily to the reader can only help it. Its conditional mean is respectively

$$\mu_j(\theta) = \theta(1 - 3q_j/2), \qquad \mu_j(\theta) = -30q_j + \theta(1 + 3q_j/2). \tag{9}$$

For either allowed menu the derivative $a_j = \partial_\theta \mu_j$ lies in $[-1/2, 1/2]$.

Fix the independent random seed. At a given history the selection rule for q_j has no additional dependence on θ. Hence the conditional Gaussian score is $a_j(Y_j - \mu_j)$ even when the rule is adaptive. These scores have conditional mean zero and conditional second moment $a_j^2 \leq 1/4$. Their cross terms vanish by conditioning on the earlier history. The score ℓ' of the complete transcript therefore obeys $\mathbb{E}_\theta(\ell')^2 \leq N/4$. A stopped protocol can be padded to N reads that the estimator ignores; averaging over the independent seed preserves the bound.

Use the prior

$$\pi(\theta) = \frac{2}{20}\sin^2\frac{\pi\theta}{20}, \qquad I_\pi = \int_0^{20} \frac{(\pi')^2}{\pi} = \frac{4\pi^2}{20^2}. \tag{10}$$

Clip the estimator to $[0, 20]$, which cannot increase its risk. The prior vanishes at both endpoints. Integration by parts in the joint prior–transcript density gives

$$\mathbb{E}\big[(\widehat{\theta} - \theta)(\pi'/\pi + \ell')\big] = 1. \tag{11}$$

The clipped error is bounded, the prior Fisher information is finite, and the Gaussian score has the square-integrable bound above; the integration and boundary terms are consequently justified. The prior–likelihood cross term is zero because every likelihood score has mean zero. Cauchy–Schwarz yields a Bayes risk of at least $(I_\pi + N/4)^{-1}$, and worst-case risk dominates Bayes risk. This is the van Trees argument [3, 4], applied to the permitted adaptive directional experiment. □

The lower bound does not compare null data with naive volume sampling. Both non-null readers know the two-parameter source law, can cancel its trace by combining measurements, and may allocate their reads after seeing earlier results. Their disadvantage is the restricted gain for separating θ from α in one causal type.

Proposition 3.3 (Finite-record separation)**.** *For the common nine-bit readout,*

$$C_{\mathrm{N}} = 1, \qquad 3 \le C_{\mathrm{T}}, C_{\mathrm{S}} \le 8. \tag{12}$$

With four reads the null protocol has worst-case risk less than 0.563*, whereas either non-null menu has risk at least* 0.9101698*.*

Proof. Let $\varphi(a)$ and $\overline{\Phi}(a)$ be the standard normal density and upper tail, and put

$$B(a) = 2\{a\varphi(a) + \overline{\Phi}(a)\}. \tag{13}$$

For a signal of absolute value at most $64 - a$, clipping contributes an L^2 error at most $\sqrt{B(a)}$: its squared error is bounded by $Z^2\mathbf{1}_{\{|Z|>a\}}$, whose expectation is (13). Rounding adds at most $\Delta/2 = 1/4$. Thus the per-record digitization error has L^2 norm at most $1/4 + \sqrt{B(a)}$.

At $q = 0$, $a = 44$ works for every source in $\mathcal{B}$. The clipped average of N digitized null records therefore has risk at most

$$\left(N^{-1/2} + \tfrac{1}{4} + \sqrt{B(44)}\right)^2. \tag{14}$$

This follows by Minkowski from the undigitized sample mean; it requires no assumption that rounding errors are unbiased or independent. For $N = 1$ the bound is less than $1.563 < 1.6$; for $N = 4$ it is less than 0.563. Without any record the minimax risk on $\theta \in [0, 20]$ is 100, attained by the constant estimate 10. A random estimate cannot reduce this value. Hence $C_{\mathrm{N}} = 1$.

For $N = 2$, (8) is $1.6702966555\ldots > 1.6$; for $N = 4$ it is $0.9101698376\ldots$. To prove finite non-null sufficiency, use six reads at $q = 1/3$ and two at $q = 1$ for the spacelike menu, or six at $q = -1/3$ and two at $q = -1$ for the timelike menu. In both cases

$$\widehat{\theta}_{\mathrm{an}} = \tfrac{3}{2}\overline{Y}_{\mathrm{low}} - \tfrac{1}{2}\overline{Y}_{\mathrm{high}} \tag{15}$$

cancels α and has variance $1/2$. All these signals have clipping margin at least 14. Replace each average by its digitized average and clip the final estimate to $[0, 20]$. Its risk is at most

$$\left(\sqrt{\tfrac{1}{2}} + \tfrac{1}{2} + 2\sqrt{B(14)}\right)^2 < 1.458 < 1.6. \tag{16}$$

Here $B(14) = 3.08608544\ldots \times 10^{-42}$ and $B(44) = 1.40785656\ldots \times 10^{-419}$. This completes the comparison. □

The one-record datum is minimal under deletion of a complete read: its only proper subdatum in that menu is no record. Nine bits have not been proved to be the minimal code length. Further coarsening may still meet the error target. Likewise, (12) bounds the non-null optimum rather than identifying whether three through seven reads suffice.

The read-cost ratio is now well defined:

$$\rho_{\varepsilon_*} = \frac{C_{\mathrm{N}}}{\min(C_{\mathrm{T}}, C_{\mathrm{S}})}, \qquad \frac{1}{8} \le \rho_{\varepsilon_*} \le \frac{1}{3} < 1. \tag{17}$$

It prices actual uses of the common interface. It is not a ratio of total apparatus energies, preparation costs or optimal detector work.

4 What produces the advantage, and what removes it

The causal distinction acts through a concrete statistical geometry. Null contraction annihilates the entire line of tensors αg. Within either finite-boost non-null menu, the two coefficients are only partly separable at fixed gain. This explanation is stronger than calling the boundary thin, but also more conditional: a detector menu determines which alternatives the word "cheapest" covers.

4.1 The analogue benchmark

There is a sharp benchmark if the source range is allowed to grow and the readout is analogue. Let $\theta \geq 0$, $\alpha \leq 0$ and $|q| \in [q_0, 1]$ within one causal type, with $0 < q_0 < 1$. Replace the compact prior interval by $[0, L]$. The least-favourable families are $\alpha = -2\theta/(1+q_0)$ for spacelike directions and $\alpha = -2L/(1+q_0) + 2\theta/(1+q_0)$ for timelike directions. Their gains are bounded by $r = (1-q_0)/(1+q_0)$. Sending L to infinity gives

$$\inf_{\text{protocols}} \sup_{\theta \geq 0, \alpha \leq 0} \mathbb{E}(\widehat{\theta} - \theta)^2 \geq \frac{(1+q_0)^2}{(1-q_0)^2} \frac{\sigma^2}{N}. \tag{18}$$

For $q_0 = 1/3$ and N divisible by four, allocate $3N/4$ reads to the small-magnitude endpoint and $N/4$ to the other. Formula (15) attains $4\sigma^2/N$, so the lower bound is exact in that analogue benchmark. Null averaging attains σ^2/N, with the matching location-model lower bound obtained from the same priors. The growing-box argument is an ideal statistical benchmark, not an additional claim about a bounded-energy laboratory.

4.2 Mixed types and near-null directions

Allowing both $q = 1$ and $q = -1$ changes the permission. Equal allocation and the mean of the two group averages cancel α and give variance σ^2/N, equal to null averaging for even N. A physical front end that measures the sum of these contractions directly is another change of interface. Neither is included in the single-type lower bound.

The boost restriction is also essential. A one-read non-null estimate with small $|q|$ has, on $\mathcal{B}$, digitized root-mean-square error bounded by

$$1 + \tfrac{1}{4} + 30|q| + \sqrt{B(14)}. \tag{19}$$

For $|q| \leq 0.0004$ its squared bound is below 1.6. Thus sufficiently near-null non-null probes meet the same one-record task. There is no discontinuous universal advantage attached merely to the sign of $g(w, w)$. The finite separation is between specified implementable menus, while nullity gives exact nuisance rejection.

4.3 The source that the null record omits

At fixed θ, every value of α gives exactly the same null output law, for any number of reads. Consider instead estimating the energy density $S_{00} = \theta - \alpha$. The sources $\alpha = 0$ and $\alpha = -30$ differ in energy density by 30 but are indistinguishable to this interface. No nonlinear processing, additional null samples or more precise null digitizer separates them. Their common output forces worst-case absolute energy error at least 15, and worst-case squared error at least 225 for those two sources.

This is a blind direction, not a gravitational on/off rule. A detector that rejects a trace contribution does not remove that contribution from the matter action. To obtain a complete source certificate one must add a trace-sensitive measurement, calibrate the missing term from independent data, or bound it sufficiently for the response task. The companion analysis [20] asks exactly when a readout supports such a certificate. The two responsibilities fit together: economical task readout and complete physical source accounting need not select the same observable algebra.

5 Reading the established null cluster

The source task isolates one mechanism. The wider cluster contains others, whose distinct assumptions must remain visible. Their conjunction suggests a research direction rather than a common derivation already in hand.

5.1 Causal transport and characteristic data

A relativistic null cone bounds causal influence; it does not confine every Green function to the cone. Characteristic reconstruction needs the field equation, gauge constraints and suitable supplementary data [1]. Massive fields and curved backgrounds can have tails. This mechanism removes redundant acquisition using a propagation law, whereas (3) rejects a nuisance at the detector. Combining them requires one detector and reconstruction analysis.

5.2 Restricted access, thermality and entropy

Hawking and Unruh thermality depend on the state, geometry and observation regime [5, 6]. The vacuum wedge has geometric modular flow in the Bisognano–Wichmann setting [12, 22]; an arbitrary restriction need not [13]. Horizon entropy, light-sheet bounds and quantum null energy inequalities relate particular causal boundaries to information [7, 8, 10]. Quantum focusing remains a conjecture outside its proved regimes [9]. Neither these results nor Jacobson's thermodynamic argument [11] identify finite detector cost with area. Global entropy, restricted-state entropy and acquired information remain different quantities.

5.3 Asymptotic memory and symmetry

Bondi–Sachs radiation, soft relations and memory connect null-infinity data to asymptotic charges and persistent detector changes [14, 15, 16, 17]. Radiative records can retain a travelling source economically, while constraint, Coulombic or trace data may still be needed for a complete source certificate. A retained displacement is an imprint of the passage, not by itself an observer's self-description.

5.4 Massless propagation and the finite-resolution edge

Null momentum characterizes relativistic massless dispersion; gauge structure can protect masslessness in a specified theory, but does not follow from nullness alone. The mass–energy companion distinguishes spectral mass from mass inferred through a readout dictionary [21]. Finite resolution near the cone and averaging momenta rather than invariant masses can change that reading. Particle masses and mixing still require their dynamical model.

6 A stronger programme and its tests

The finite experiment supplies an answer to a restricted but substantive question: given this source, this target and these actual detector menus, null data are sufficient and have a strict read-count advantage. The larger explanatory claim is that related efficiencies help account for the recurrence of null structure in physical descriptions. It can be formulated without making null data universally complete.

Conjecture 6.1 (Task-adapted null interfaces)**.** For important radiation and causal-response tasks, there are physically implementable null interfaces that attain lower registered resource cost than the admissible non-null interfaces at the same task error, because they combine characteristic transport with rejection of source components irrelevant to that task. The cost advantage survives a common accounting of detector preparation, alignment, bandwidth, clocks, computation and any source-completion data required by the target.

The conjecture has two mechanisms and two ways to fail. Characteristic transport can reduce redundant acquisition, but tails or unmeasured constraint data can make the chosen boundary inadequate. Nuisance rejection can improve conditioning, but a mixed or near-null competitor can match it. Adding realistic direction-setting and meter-renewal costs could also overturn a sample advantage. These are quantitative competing explanations, not exceptions to be excluded after a comparison.

A next experiment should use one field preparation, one causal response target and a common menu of finite surface detectors. Every reconstruction would receive the same propagation law. The task could be a specified outgoing waveform or a smeared response, rather than recovery of all bulk degrees of freedom. The null and non-null reconstructions would be judged by the same loss and the same resource vector. Their advantage would then be attributed separately to propagation redundancy and to coupling conditioning. An interacting background or a field with tails would test whether source-completion data erase the saving. This extends the present mechanism rather than promoting it to a theorem about all hypersurfaces.

An interface can be sufficient for one action and incomplete for another. The null record estimates θ efficiently; its blindness to α matters when the task becomes total source energy. Thus causal records, accessible correlations and complete physical response need not select the same data. Q-RAIF's No View analysis makes this source distinction explicit [20]. A trace term unseen by a null instrument is not thereby removed from gravity. Applications to vacuum energy, black-hole information or dark sectors require their own source and retrieval predictions, beyond (12).

7 Conclusion

Null contraction can make a physical task cheaper by discarding a source component that the task does not need. We exhibited the effect with the same scalar-source family, the same actual pointer law and the same finite digitizer for all competitors. The proof covers adaptive non-null readers and records both a sufficient null protocol and finite sufficient non-null protocols. Its advantage depends on the declared direction permission and is lost in relevant relaxed menus.

The omitted trace contribution supplies the counterweight: an economical null interface is not automatically a complete interface to matter or gravity. The wider null cluster is most promising where causal transport, restricted access and task-relevant source reduction reinforce one another. Establishing that conjunction in a common field experiment, including its completion and renewal costs, is the next test of the programme.

References

[1] R. M. Wald, *General Relativity*, University of Chicago Press (1984).

[2] W. Hu, *Astro 408: Cosmological Perturbation Theory*, lecture set 2, scalar-field slides, University of Chicago (2018), `https://background.uchicago.edu/~whu/Courses/Ast408_18/ast4PT_02.pdf`.

[3] R. D. Gill and B. Y. Levit, Applications of the van Trees inequality: a Bayesian Cramér–Rao bound, *Bernoulli* **1**(1/2), 59–79 (1995).

[4] E. Gassiat, D. Pollard and G. Stoltz, *Revisiting the van Trees inequality in the spirit of Hájek and Le Cam*, research draft, Sections 4 and 7.2, `https://stoltz.perso.math.cnrs.fr/Publications/vanTrees-LeCam-TempWeb.pdf`.

[5] S. W. Hawking, Particle creation by black holes, *Communications in Mathematical Physics* **43**, 199–220 (1975).

[6] W. G. Unruh, Notes on black-hole evaporation, *Physical Review D* **14**, 870–892 (1976).

[7] J. D. Bekenstein, Black holes and entropy, *Physical Review D* **7**, 2333–2346 (1973).

[8] R. Bousso, A covariant entropy conjecture, *Journal of High Energy Physics* **07**, 004 (1999).

[9] R. Bousso, Z. Fisher, S. Leichenauer and A. C. Wall, Quantum focusing conjecture, *Physical Review D* **93**, 064044 (2016).

[10] R. Bousso, Z. Fisher, J. Koeller, S. Leichenauer and A. C. Wall, Proof of the quantum null energy condition, *Physical Review D* **93**, 024017 (2016).

[11] T. Jacobson, Thermodynamics of spacetime: The Einstein equation of state, *Physical Review Letters* **75**, 1260–1263 (1995).

[12] J. J. Bisognano and E. H. Wichmann, On the duality condition for a Hermitian scalar field, *Journal of Mathematical Physics* **16**, 985–1007 (1975).

[13] R. Haag, *Local Quantum Physics: Fields, Particles, Algebras*, Springer (1996).

[14] H. Bondi, M. G. J. van der Burg and A. W. K. Metzner, Gravitational waves in general relativity VII, *Proceedings of the Royal Society A* **269**, 21–52 (1962).

[15] R. K. Sachs, Gravitational waves in general relativity VIII, *Proceedings of the Royal Society A* **270**, 103–126 (1962).

[16] S. Weinberg, Infrared photons and gravitons, *Physical Review* **140**, B516–B524 (1965).

[17] A. Strominger, *Lectures on the Infrared Structure of Gravity and Gauge Theory*, Princeton University Press (2018).

[18] S. Liu, *Lorentzian Read-out from Local Unitary Propagation: Clifford Relations, Causal Cones and Spectral Mass*, 2026; series record DOI: 10.5281/zenodo.18525876.

[19] S. Liu, *Representing Control in the World: Retained History, Physical Sources and Renewal in Q-RAIF*, 2026; series record DOI: 10.5281/zenodo.18528934.

[20] S. Liu, *No View from Nowhere: Source Selection, Physical Reaction and Read-out Closure in Q-RAIF*, 2026; series record DOI: 10.5281/zenodo.20284454.

[21] S. Liu, *Recovering Mass–Energy Equivalence within Algebraic Lorentzian Read-out: Spectral Mass, Lattice Windows and Finite Records*, 2026; series record DOI: 10.5281/zenodo.20280951.

[22] J. J. Bisognano and E. H. Wichmann, *On the duality condition for quantum fields*, J. Math. Phys. **17**, 303–321 (1976), doi:10.1063/1.522898.